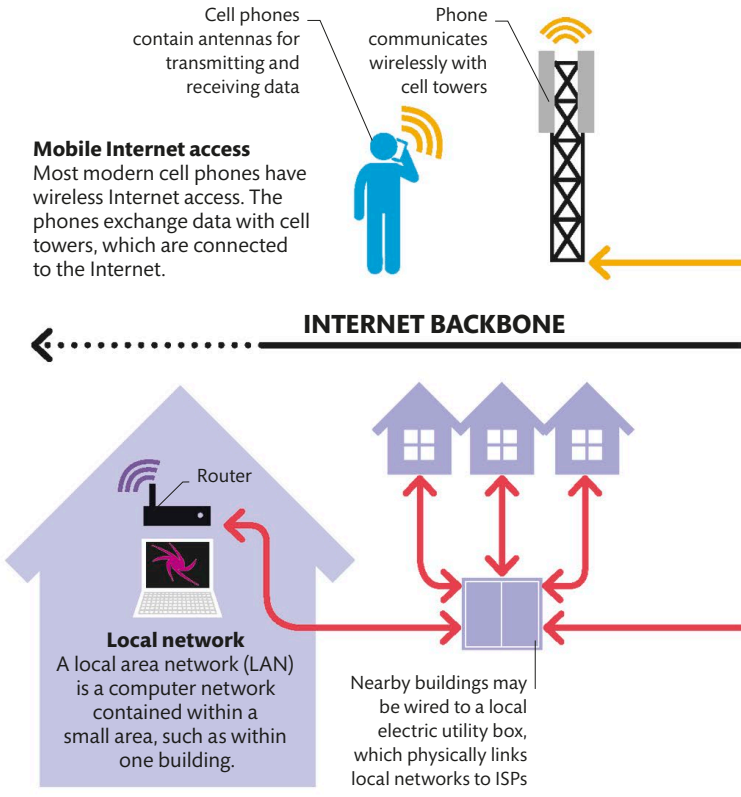
The Internet

The Internet is a global network of connected computers that exchange data using a common set of rules. It supports important applications such as email and the World Wide Web.

A network of computers

Users can access the Internet from an end point such as a smartphone or a computer. These devices are normally connected to the Internet through an Internet service provider (ISP), which adds them to its network and also assigns a unique reference (called an IP address) to each device. These networks are in turn connected to other networks to form larger networks. The Internet is the collection of all these interlinked networks of computers, meaning that any computer on the Internet can connect to any other. When computers exchange data, layers of software manage the process that divides data into packets, which travel through routes of wires, fiber-optic cables, and wireless connections to reach their end destination.



Routes for data

Older telecommunications networks rely on circuit switching to send and receive data, meaning that direct, wired connections are formed between end points during the exchange. Nowadays, packet switching is the primary method for exchanging data online. Software splits data up into sections called packets, which are labeled with their destination IP address and instructions for reassembly. These packets are directed to their end point via different routes and then reassembled at their destination. Packet switching allows for communication channels to be used far more efficiently, because different sets of data can travel through them at the same time.

